Exponential regression

1

a 3

b

i Yes ii No iii No iv No

2

a 4

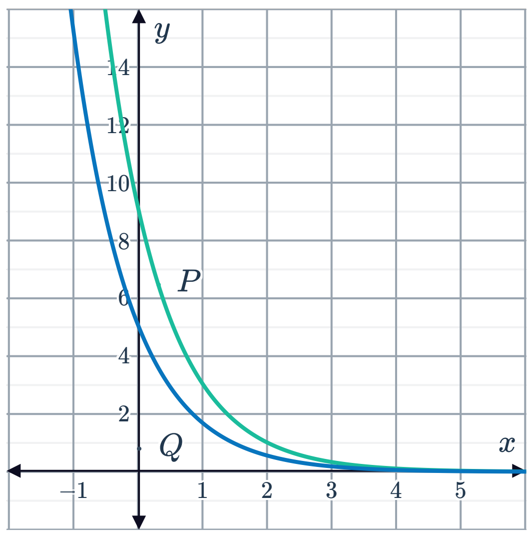
b

i No ii Yes iii No iv No

3

a Decreasing

b



c Both functions aprroach 0.

d

i A vertical dilation by a factor of 9.

ii A vertical dilation by a factor of 5.

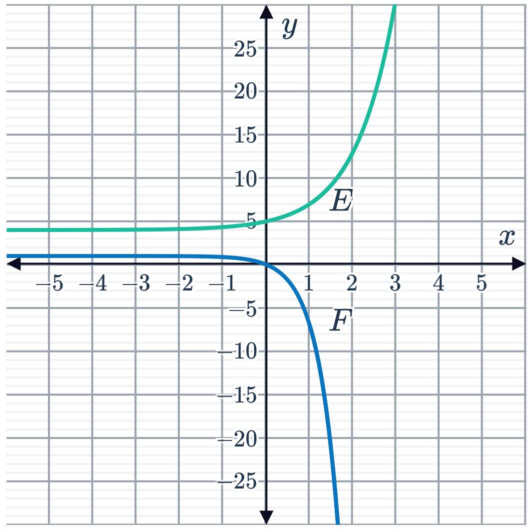
4

a Once

b Function P

5

a



b Zero times

c As $x \to \infty$, the function $E \to \infty$ and function $F \to -\infty$.

Applications

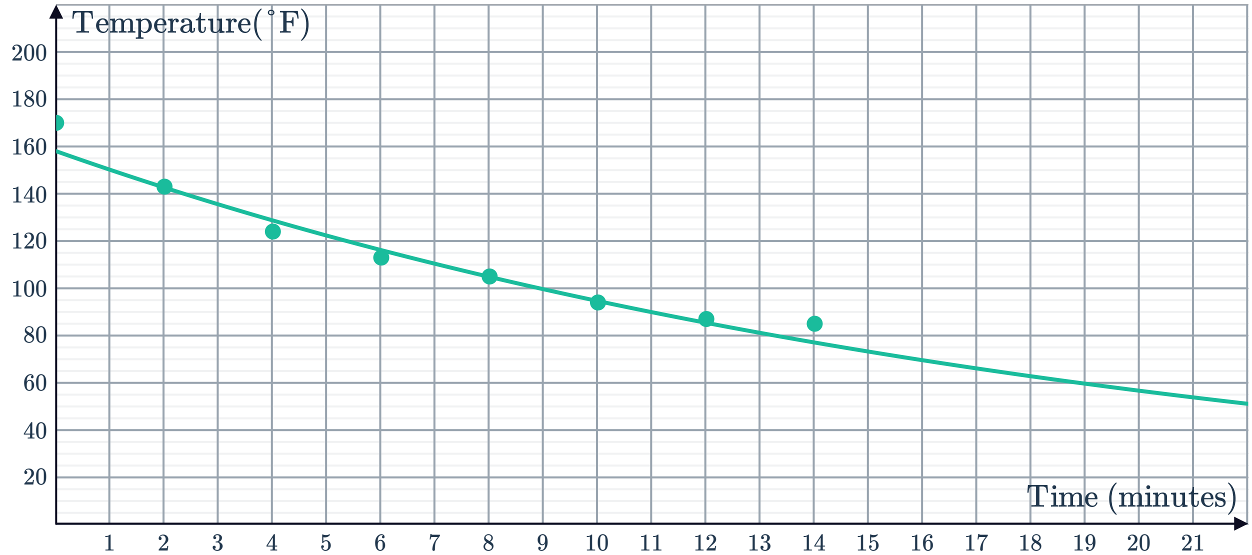
6

a We cannot approximate the new pan's quality rating by fitting a linear, exponential or quadratic function to the data.

b False

7

a



Yes, an exponential model is appropriate because when drawn on a graph it has an exponential shape which is decreasing less rapidly over time.
Also, if we calculate an exponential regression on this data set using the exponential function, we get $R^2 = 0.9647$ which is very close to 1, so there is a strong correlation to the exponential model suggesting an exponential association.

b $y = 157.9684\,(0.9521^x)$, where x is the number of minutes and y is temperature.

c Most of the points corresponding to actual measurements lie on the graph of the function.

d $59.2\,^\circ\,\mathrm{F}$

8

a C

b

t	0	10
y	0	49 000

c $2\,439\,000$

9

a $M = 50\,000\,(1.032)^t$

b $P = 49\,000\,(1.061)^t$

c $\$9500.24$

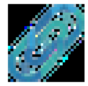
d Elizabeth's

10

a 8 million

b 7130 million

c 0.87%



11

a 10

b $20\,\%$

c $100\,\%$

d The population of bacteria in the dark environment is increasing at a faster rate than the population of bacteria in the sunlit environment.

12

a $A = P\,(1.053)^t$ b $B = P\,(1.01)^{4t}$ c Bank A

13

a The sale price of property. b 5.5%

c -3.6%

14

a $P = 90 + 5t$ b $Q = 90 \times (1.04)^t$

c the current company d opening his own business

15

a $P = (0.85)^t$ b Maximilian c C d 13%

e No

16

a $P = 2\,410\,000\,(1.022)^n$

b $F = 3\,374\,000\,(1.02)^n$

c Yes

d The food production is increasing, but at a slower rate than the population's rate of increase.

e Over a decade from now.

17

a 179

b 132

c person B's

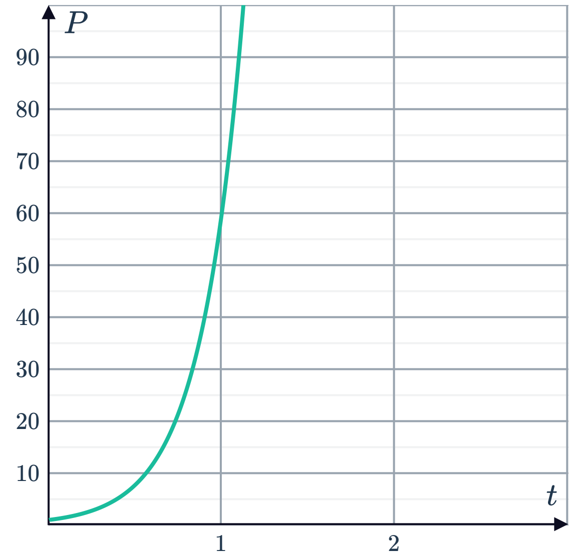
d 0.7

e if you have a high sugar diet, your blood sugar level peaks at greater amounts and decreases more rapidly than a low sugar diet

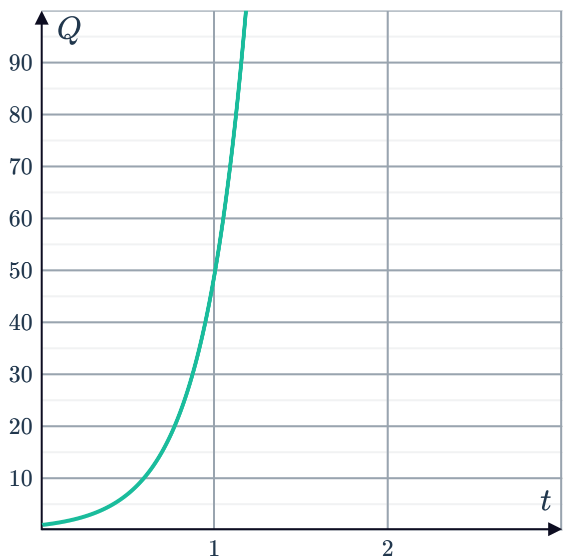
18

a Faster

b



c



d No